

WFH Burnout Predictor

Machine
Learning Project
A.Y. 25-26

Predicting burnout score from daily work-from-home signals

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Machine Learning and Data Mining, Module 2

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Project question

Can daily work-from-home indicators predict burnout score for employees not seen during training?

Main task

Regression on burnout_score.

Secondary task

Classification of burnout_risk, used mainly for interpretation.

Focus

The dataset is simple, so the analysis emphasizes data and result understanding.

A clean dataset with repeated employee observations

The dataset contains daily observations from work-from-home employees. Each user appears 10 times, so the identifier is useful for evaluation but not as a predictive feature.

1,800

observations

180

users

10

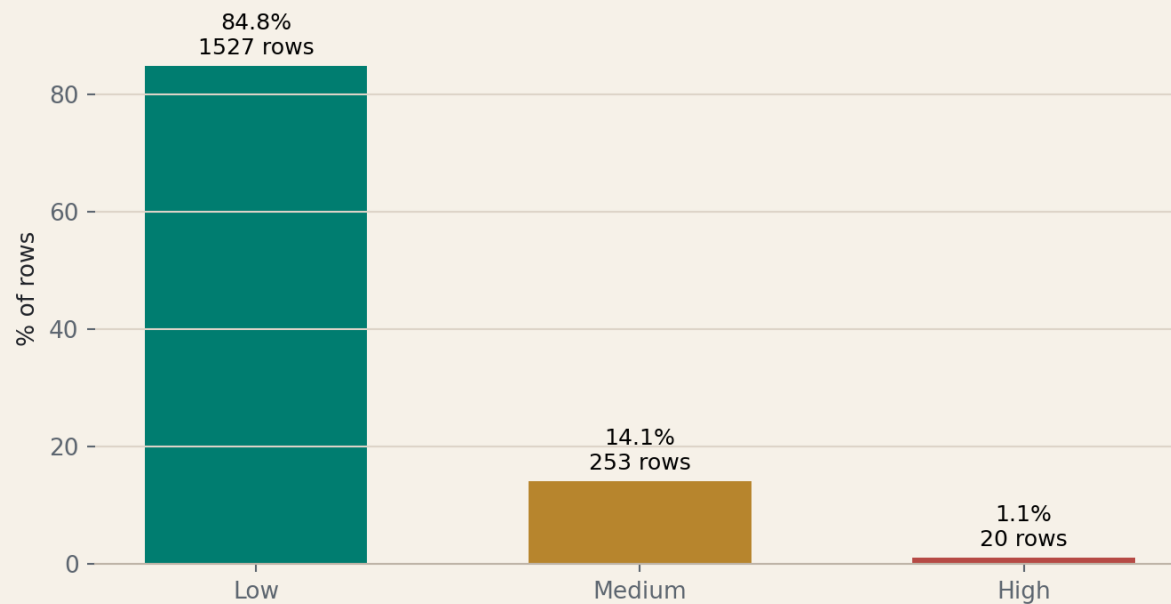
records per user

0

missing values

- Predictors include work hours, screen time, meetings, breaks, after-hours work, sleep and task completion.
- Targets are burnout_score and burnout_risk.
- The technical quality is high, but external representativeness remains unknown.

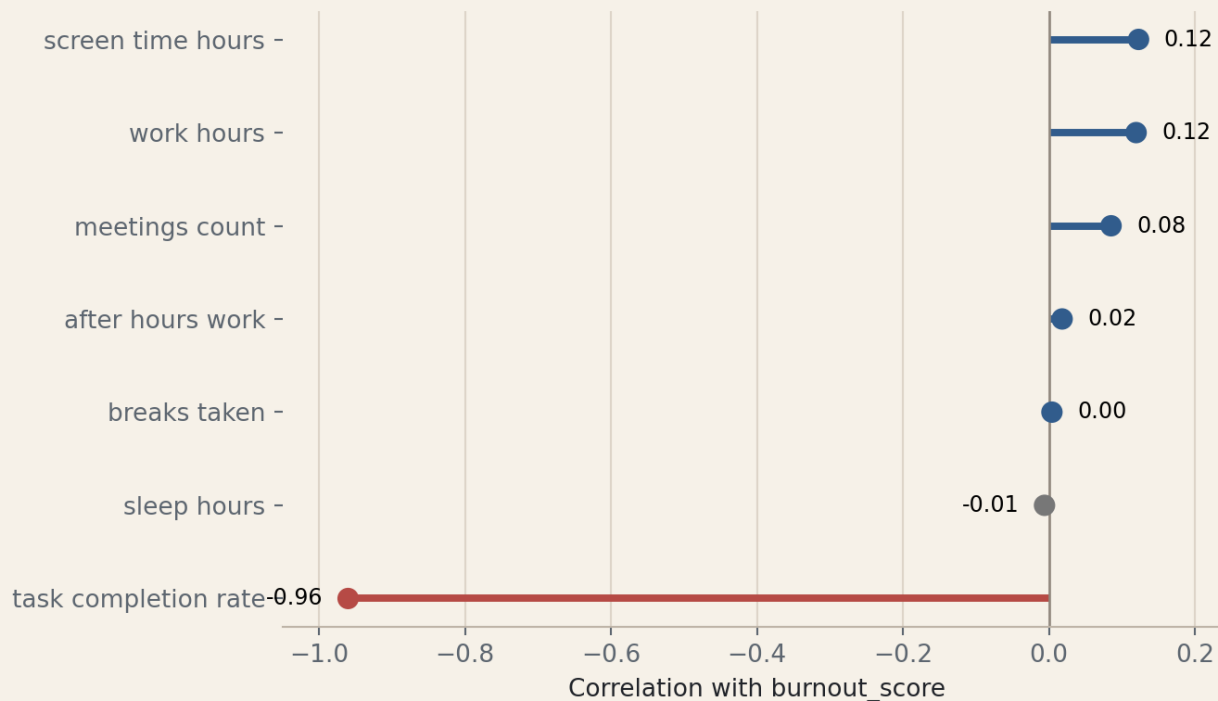
The risk label is highly imbalanced



Why this matters

- Low-risk rows dominate the dataset.
- High risk appears in only 20 records.
- A classifier can look accurate while missing the rarest class.
- The continuous score is therefore the main target.

Most of the signal comes from one feature



task_completion_rate has about **-0.96** correlation with **burnout_score**.

This is useful for prediction, but it is also the main warning sign: the feature may be close to the target-generation logic.

The split is designed around users, not rows

Excluded

user_id is not used as a predictor.

Grouped split

records from the same employee stay either in train or in test.

Pipeline

encoding, scaling and model fitting are handled inside scikit-learn pipelines.

1,440 / 360

train/test rows

144 / 36

train/test users

0

overlapping users

Baselines first, then model comparison

Baseline

Dummy models define the minimum acceptable result.

Linear

Ridge checks whether the signal is mostly linear.

Non-linear

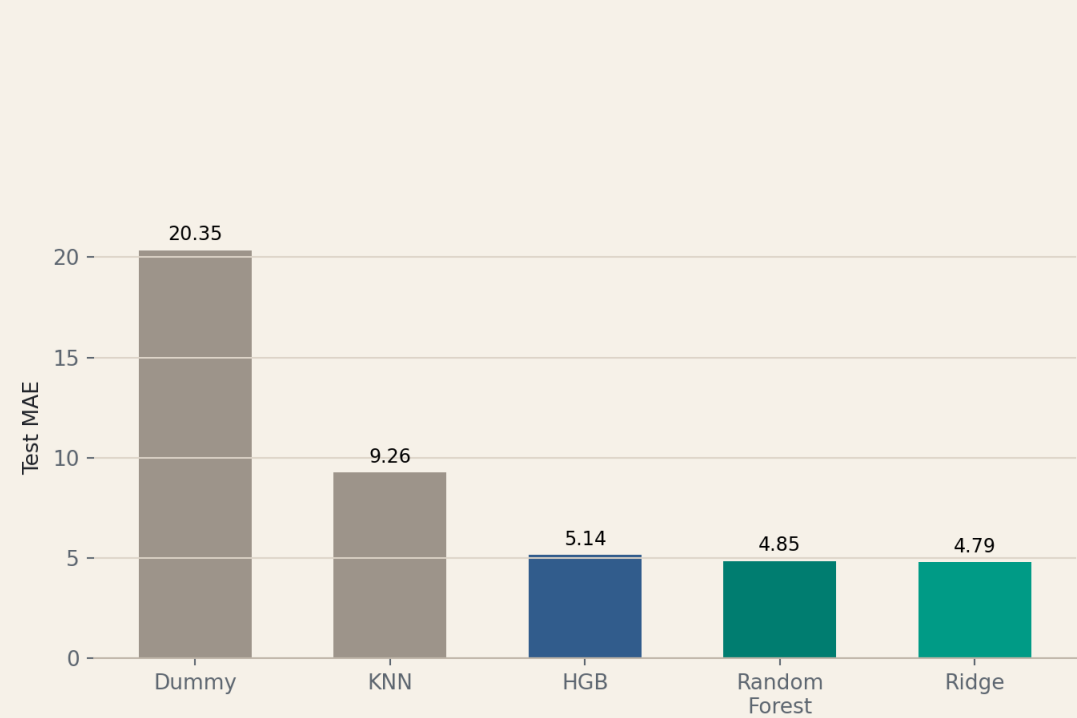
Random Forest and Gradient Boosting capture interactions.

Validation

GroupKFold keeps the user-level evaluation logic during selection.

Model	CV MAE	CV R2
Random Forest	4.97	0.923
Ridge	5.06	0.916
Hist Gradient Boosting	5.19	0.917
KNN	9.03	0.744
Dummy median	18.46	-0.046

All meaningful models beat the dummy baseline

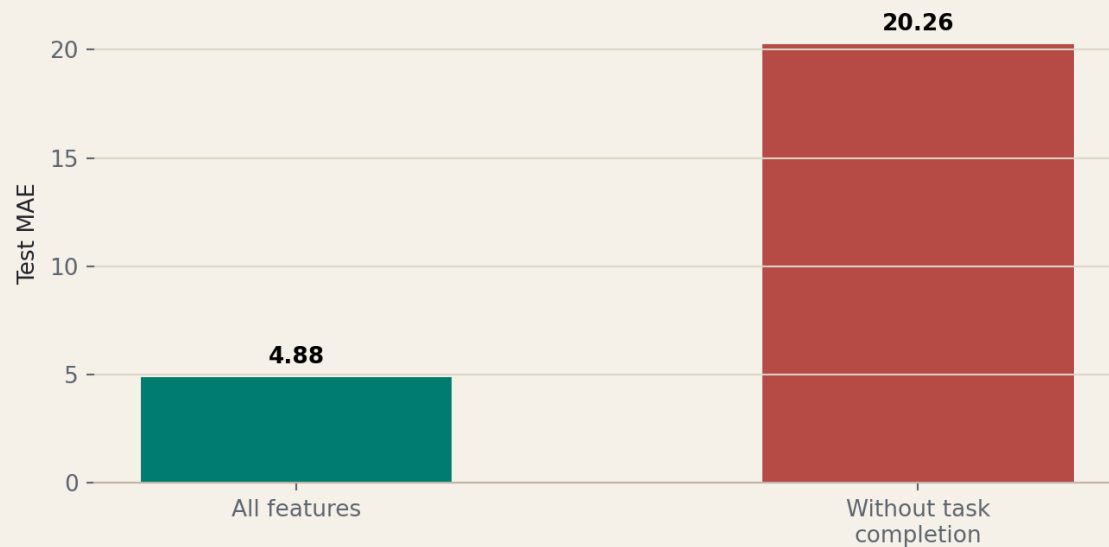


Model	MAE	RMSE	R2
Ridge	4.79	6.31	0.934
Random Forest	4.85	6.41	0.932
Hist Gradient Boosting	5.14	6.73	0.925
KNN	9.26	11.65	0.776
Dummy median	20.35	25.80	-0.099

4.88
tuned Random Forest MAE

0.936
tuned Random Forest R2

High performance depends on task completion



Removing `task_completion_rate` moves performance close to the dummy baseline.

This does not invalidate the pipeline. It narrows the claim: the project demonstrates predictive modeling and exposes where the model's reliability comes from.

The risk labels are informative but limited

Class	Precision	Recall	F1	Support
Low	0.99	0.93	0.96	294
Medium	0.68	0.73	0.70	62
High	0.17	0.75	0.27	4

Best classifier: balanced Logistic Regression

Balanced accuracy 0.803, macro F1 0.646. High-risk recall is 0.75, but only 4 High examples are present in the test set.

The model is useful only in a cautious decision context

Interpretability

Permutation importance confirms that `task_completion_rate` dominates the prediction.

Fairness checks

Errors are inspected across operational groups such as `day_type`, `after_hours_work` and work-hour bands.

Deployment

Use as aggregate decision support only. Automatic individual decisions are out of scope.

External validation is required before applying the pipeline to a real organization.

Key insights & conclusions

- **Data understanding drives the project:** the dataset is clean but simple, with repeated users and a rare High-risk class.
- **The evaluation is conservative:** users are grouped to avoid optimistic train/test leakage.
- **The regression result is strong:** tuned Random Forest reaches MAE 4.88 and R2 0.936 on unseen users.
- **The main limitation is explicit:** performance depends strongly on task_completion_rate.
- **Final interpretation:** a transparent predictive pipeline, not evidence of causal burnout mechanisms.